Topic Modeling and Extractive Summarization on CNN/Daily Mail Dataset

A Text Mining Project

Stephen Adu Poku Yeboah - Emmanuel Ampah
Student ID: 852121 - 868745

Academic Year 2025/2026

Abstract

This project investigates unsupervised topic modeling and extractive text summarization techniques on the CNN/Daily Mail news corpus. For topic discovery, probabilistic (Latent Dirichlet Allocation), linear algebraic (Latent Semantic Analysis), and embedding based (BERTopic) approaches are systematically compared using coherence metrics and qualitative inspection. For summarization, two extractive methods TextRank with TF–IDF weighting and centroid-based ranking using Word2Vec embeddings are implemented and evaluated using ROUGE scores against human-written highlights. Extensive preprocessing and hyperparameter tuning are conducted to ensure fair comparison. Results show that LDA with a small number of topics captures dominant editorial themes most coherently, while BERTopic excels at fine-grained, event-level topic discovery. In summarization, TF–IDF-weighted embeddings improve sentence salience estimation, producing summaries that better align with reference highlights. These findings highlight the complementary strengths of probabilistic and embedding-based methods for large-scale news analysis.

1 Introduction

1.1 Problem Description

This project addresses the challenge of automatically analyzing and processing large-scale news articles from the CNN/Daily Mail dataset. With the exponential growth of digital news content, efficient automated techniques for topic discovery and content summarization have become essential for information management and retrieval.

1.2 Tasks Addressed

We address two core text mining tasks:

1. **Topic Modeling:** Topic Modeling is an unsupervised machine learning technique that involves scanning a set of documents to detect word and phrase patterns within

them. The goal is to automatically cluster word groups and similar expressions that best characterize a given set of documents. It provides collections of words that cohesively make sense together, representing what we interpret as topics. Discovering latent themes within the news corpus using unsupervised learning techniques (LDA, LSA, BERTopic). Each word in the cluster have a probability of occurrence for the given topic, dierent topics have their respective clusters of words along with corresponding probabilities, and dierent topics may share some words and a document can have more than one topic associated with it

2. **Text Summarization:** Generating concise extractive summaries using graph-based ranking (TextRank with TF-IDF) and embedding-based approaches (Word2Vec with TF-IDF weighting). Text summarization is the process of distilling the most important information from a source (or sources) to produce an abridged version for a particular user (or users) and task (or tasks). The main goal is to produce an abridged version of a text that contains information that is important or relevant to a user. Text summarization can be applied in all the task that involve summaries, such as headlines or reviews. Applying summarization to a text or a document have many advantages, because summaries reduce reading time, so when searching for documents, summaries make the selection process easier. Automatic summarization improves the eectiveness of indexing (IR) and other TMtasks (e.g., classification, clustering...), and the algorithms are less biased than human summarizers. Personalized summaries are useful in question-answering systems as they provide personalised information. Using automatic or semi-automatic summarization systems enables commercial abstract services to increase the number of text documents they can process.Text summarization have many dimensions:

   (a) **B**ased on input type: Single-document, Multi-document

   (b) **B**ased on the purpose: Generic, Domain-specific, Query-based.

   (c) **B**ased on output type: Extractive, Abstractive

   We adopted the Output Type dimension using the Extractive method. Extractive summarization is still used today and remains a relevant and widely employed approach in Natural Language Processing and Text Mining. The main advantages like the fact that it preserves of source content, since it selects sentences directly from the source document, extractive summarization inherently preserves the language and content used in the original text. Or again it Reduces risk of generating incoherent text, by avoiding the challenge of generating new sentences, reducing the risk of generating grammatically incorrect or semantically inconsistent text. Finally it remain interpretable, because the extracted sentences are often directly interpretable and traceable back to the source material, making it clear which parts of the document contributed to the summary.

1.3 Approach Overview

Our approach involves comprehensive text preprocessing with hyperparameter tuning, implementation of multiple algorithms for each task, and systematic evaluation using coherence metrics (topic modeling) and ROUGE scores (summarization).

2 Dataset

2.1 CNN/Daily Mail Dataset

The CNN/Daily Mail dataset [1] is a widely-used benchmark containing news articles paired with multi-sentence summaries (highlights). Each article includes human-written highlights that serve as reference summaries for evaluation.

2.2 Dataset Statistics

Table 1: Dataset Statistics

Statistic	**Value**
Total Documents	92,579
Average Tokens per Document	312.7
Min/Max Tokens	4 / 1,334
Original Vocabulary Size	310,324
Filtered Vocabulary Size	10,000
Average Highlights per Article	3-4

2.3 Dataset Characteristics

The corpus covers diverse news topics including politics, international affairs, sports, entertainment, and human interest stories. Each story file contains the article text followed by @highlight markers separating the reference summary sentences.

3 Methods

3.1 Text Preprocessing

3.1.1 General Pipeline

Topic modeling methods generally demand extensive preprocessing to ensure topic coherence and interpretability., and rely on word co-occurrence patterns to discover semantic structures. So, they are sensitive to lexical noise. Whereas text summarization can operate effectively with minimal preprocessing. Our preprocessing pipeline includes: (1) whitespace and special character normalization, (2) case folding, (3) punctuation and number removal, (4) tokenization, (5) stop word removal (314 words: 198 base + 117 custom), and (6) lemmatization.

3.1.2 Bigram Detection

We employed Gensim's Phrases model to detect meaningful word pairs on 50,000 documents. Table 2 shows the hyperparameter tuning results.

Table 2: Bi-gram Hyperparameter Tuning Results

min_count	threshold	Phrases	Coherence
5	1.0	204,835	-2.135
10	1.0	88,374	-2.052
20	**0.5**	**41,213**	**-1.9918**

The optimal configuration identified 86,286 unique phrases including "united_states", "white_house", "north_korea", and "hong_kong".

3.1.3 Dictionary Filtering

The optimal dictionary parameters (no_below=10, no_above=0.5, keep_n=10,000) achieved 67.66% vocabulary coverage while reducing vocabulary from 310,324 to 10,000 terms.

3.2 Topic Modeling Methods

3.2.1 Latent Dirichlet Allocation (LDA)

LDA [2] is a generative probabilistic model representing documents as mixtures of topics. We tuned 84 parameter combinations including number of topics, alpha (document-topic density), eta (topic-word density), and passes.

3.2.2 Latent Semantic Analysis (LSA)

LSA applies truncated SVD to the term-document matrix to discover latent semantic dimensions. We tuned the number of topics and power iterations.

3.2.3 BERTopic

BERTopic [3] uses BERT embeddings with HDBSCAN clustering and class-based TF-IDF for fine-grained topic discovery.

3.3 Text Summarization Methods

3.3.1 Task Definition

Given a document D consisting of sentences $\{s_1, s_2, ..., s_n\}$ and reference highlights H, the goal is to select a subset of sentences that best captures the document's key information.

3.3.2 Method 1: TextRank with TF-IDF (Graph-Based)

Our first approach adapts the TextRank algorithm [4]:

1. **Preprocessed Sentence:** Using the preprocessed sentences (lowercasing, stopword removal, punctuation removal).
2. **TF-IDF Vectorization:** A TF–IDF vectorizer was used to construct bag-of-words representations for the normalized sentences. The vectorizer was configured to use inverse document frequency weighting, with no explicit minimum or maximum document frequency thresholds applied. The model was fit on the full set of normalized

sentences, and each sentence was transformed into a TF–IDF–weighted feature vector. The resulting sparse document–term matrix was subsequently converted to a dense array for downstream processing.

3. **Similarity Graph:** A graph is constructed where nodes are sentences and edge weights are cosine similarities between TF-IDF vectors. Pairwise sentence similarity was computed by taking the dot product of the TF–IDF sentence representations. This resulted in a square similarity matrix in which each entry represents the similarity between a pair of sentences in the TF–IDF feature space. The dimensionality of the matrix corresponds to the number of sentences, and values were optionally rounded for inspection and reporting.

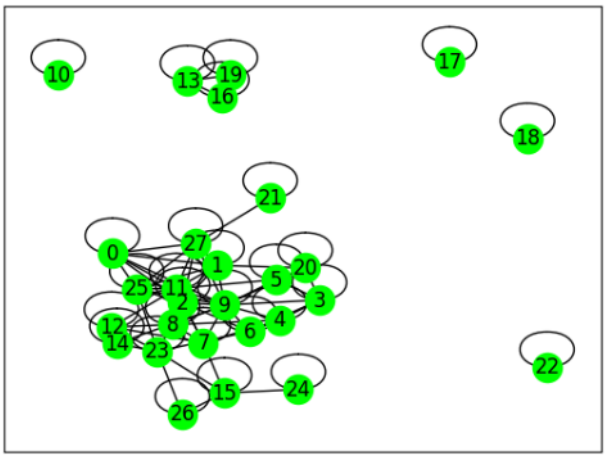


Figure 1: Example of graph similarity

4. **PageRank Scoring:** PageRank was then applied to this graph to compute an importance score for each sentence, reflecting its centrality within the similarity network.

5. **Sentence Selection:** Sentences were ranked in descending order based on their PageRank scores, and the top-ranked sentences were selected for summary construction.

The similarity matrix S is computed as:

$$S_{ij} = \frac{\mathbf{v}_i \cdot \mathbf{v}_j}{||\mathbf{v}_i|| \cdot ||\mathbf{v}_j||} \tag{1}$$

where $\mathbf{v}_i$ is the TF-IDF vector of sentence i.

```
[(0.04230836348359063, 9),
 (0.0401835106327808, 11),
 (0.039778626125533276, 15),
 (0.03950327634439168, 2),
 (0.03919316997229084, 23),
 (0.03757924594229239, 16),
 (0.03682440901369179, 8),
 (0.036164092132823726, 19),
 (0.03615556448682334, 12),
 (0.03613750108865218, 7)]
```

Figure 2: Pagerank scoring matrix

3.3.3 Method 2: Word2Vec with TF-IDF Weighting

Our second approach uses neural word embeddings with TF-IDF weighting:

All processed articles were stored in a dictionary containing three elements: (1) the original article text, (2) a list of tokenized sentences, and (3) the corresponding reference summary.

1. **Split into train and test:** To prevent information leakage, the dataset was split at the article level into training and test sets using an 80/20 split with a fixed random seed. Sentence level datasets were then derived from this article level split by aggregating the sentences belonging only to the articles in each subset. This ensured that no sentences from the same article appeared in both training and test sets. In addition to the sentence level representations, raw article texts and reference summaries were also extracted for both the training and test partitions to support modeling and evaluation.

2. **Word2Vec Training:** A Word2Vec model is trained on the corpus. Word representations were learned using a Word2Vec model (Skip-gram) trained on the tokenized training sentences only. The skip-gram architecture was used with a vector dimensionality of 150, a context window of size 8, a minimum word frequency threshold of 2, and training over 10 epochs (These parameters were obtained through a grid search). This ensured that distributional word embeddings were derived exclusively from the training data.

3. **TF-IDF Weighted Sentence Vectors:** In parallel, a TF–IDF model was fit once on the training documents. For each document, all sentence-level tokens were concatenated into a single text string, forming a corpus used to estimate inverse document frequencies. This TF–IDF model was subsequently reused for all sentence and document representations. Sentence embeddings were computed as weighted averages of Word2Vec word vectors. For each sentence, term frequencies were calculated locally, and each word embedding was weighted by the product of its term frequency and its inverse document frequency (TF–IDF). Words not present in either the Word2Vec vocabulary or the TF–IDF vocabulary were ignored. If a sentence contained no valid words, it was represented by a zero vector.

$$\mathbf{s} = \frac{\sum_{w \in s} \mathrm{idf}(w) \cdot \mathbf{e}_w}{\sum_{w \in s} \mathrm{idf}(w)} \tag{2}$$

   where $\mathbf{e}_w$ is the Word2Vec embedding and $\mathrm{idf}(w)$ is the inverse document frequency.

4. **Document Centroid:** The document representation is the mean of all sentence vectors.

5. **Centroid Scoring:** Sentences are ranked by cosine similarity to the document centroid.

6. **Selection:** Top sentences (ratio=0.4) are selected and reordered.

3.3.4 Word2Vec Hyperparameter Tuning

We performed a grid search on the train document to tune Word2Vec parameters, optimizing for average ROUGE-1 F1 score. Table 3 shows the parameter grid.

Table 3: Word2Vec Parameter Search Space

Parameter	Values
vector_size	100, 150, 200
window	5, 8, 10
epochs	10, 20
sg (Skip-gram)	1
min_count	2

3.4 Evaluation Metrics

3.4.1 Topic Modeling

- **Coherence (U_Mass):** Measures co-occurrence of top topic words.
- **Coherence (C_V):** Combines word co-occurrence with semantic similarity (higher is better).
- **Perplexity:** Measures how well the model predicts held-out documents.

3.4.2 Extractive Summarization

We use ROUGE (Recall-Oriented Understudy for Gisting Evaluation) metrics:

- **ROUGE-1:** Unigram overlap between generated and reference summaries.
- **ROUGE-2:** Bigram overlap.
- **ROUGE-L:** Longest common subsequence.

4 Experimental Evaluation

4.1 Topic Modeling Results

4.1.1 Hyperparameter Tuning

LDA tuning (84 configurations) and LSA tuning (27 configurations) were performed on 50,000 training documents with evaluation on 10,000 held-out documents.

Table 4: Best Topic Model Parameters

Model	Parameter	Value
LDA	num_topics	5
	alpha	symmetric
	eta	0.1
	passes	10
LSA	num_topics	5
	power_iters	2

Figure 3: Model comparison

4.1.2 Model Comparison

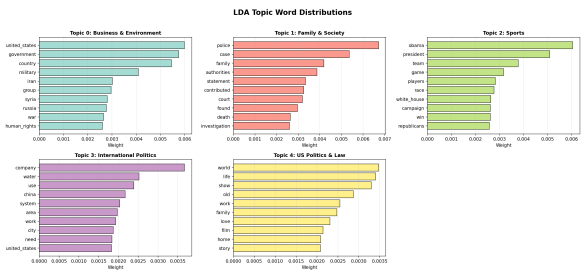
Table 5: Topic Modeling Comparison

Metric	LDA	LSA	BERTopic
Number of Topics	5	5	155
Coherence (U_Mass)	-2.1219	-2.2958	N/A
Coherence (C_V)	**0.4271**	0.3760	N/A
Perplexity	-8.4769	N/A	N/A

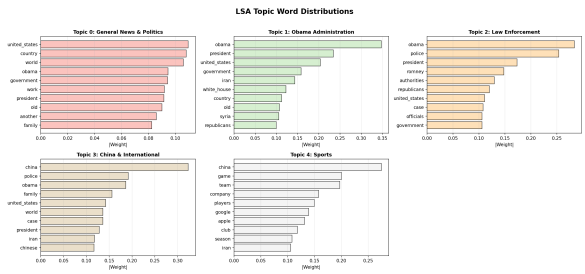
4.1.3 Discovered Topics

Table 6: LDA Topic Interpretations

ID	Label	Top Terms
0	International Politics and Conflicts	government, country, war, human$_rights, military, iran, group$
1	Crime and Law Enforcement	police, case, family, authorities, statement, court
2	US Politics and Sports	obama, president, team, game, players, race, white$_house$
3	Business and Urban Developmentl	company, water, system, area, work, city
4	Entertainment and Lifestyle	world, life, show, work, family, film



(a) LDA Topic Word Clouds



(b) LSA Topic Word Clouds

Figure 4: Topic Modeling Visualizations

4.2 Extractive Summarization

4.2.1 Word2Vec Tuning

Table 7: Word2Vec Results

vector_size	window	epochs
100	5	10
100	8	10
150	5	20
150	8	20
150	**8**	**10**

4.2.2 Summarization Method Comparison

Table 8: Summarization Results (ROUGE Scores)

Method	ROUGE-1	ROUGE-2	ROUGE-L
TextRank (TF-IDF)	0.3624	0.0896	0.1536
Word2Vec + TF-IDF	0.1933	0.0792	0.1295

4.2.3 Example Summary Graph-Based Method

Original Highlights (Reference):

"'Sao Paulo (CNN)Howś this for a miraculous outcome? A Brazilian man who was stabbed in the head during an argument survived despite the knife being lodged in his skull for over three hours.Motorcycle taxi driver Juacelo Nunes de Oliveira was at a party in a bar in the city of Agua Branca, in northeastern Piaui state, last Sunday when he got into an altercation.The 39-year-old was stabbed four times, three times in the torso, with one wound perforating a lung, and once in the head.He was first taken for treatment at a nearby hospital before being transferred to the Emergency Hospital of Teresina, a city 100 kilometers (62 miles) away – with the knife handle protruding from his head all the while."It took him nearly three hours to arrive here at the hospital," Dr. Gilberto Albuquerque, general surgeon and hospital director, told CNN on Friday."He had the knife lodged in his head for this whole period but he was not aware of it," he said. "We believe the shock and a bit of the drinking may have helped him not to understand the knife was still in his head."Pictures released by the military show Oliveira lying on a hospital trolley with what looks like a kitchen knife embedded in his skull up to the handle, just to the side of his left eyebrow.to Albuquerque, the knife stopped only when it got stuck in the manś lower jaw.Oliveira was very lucky, he said – and not just to be alive."The knife crossed his head just behind the optical nerve, got very close to the olfactory nerve and (was) just a fraction from the gustatory nerve. He could have been blinded and could have lost his senses of smell and taste, but none of that happened and he is alive and healthy."The surgery to remove the

knife was very delicate, since any slip could have breached arteries and led the patient to bleed to death, Albuquerque said.It took the hospital surgery team two and half hours to complete, and was considered to be a success."He left the hospital yesterday, five days after coming in with a knife in his head," Albuquerque said. "He is healthy and perfect and he had no ill after-effects."

Generated Summary (Graph-Based):

"a brazilian man who was stabbed in the head during an argument survived despite the knife being lodged in his skull for over three hours . " we believe the shock and a bit of the drinking may have helped him not to understand the knife was still in his head . " pictures released by the military show oliveira lying on a hospital trolley with what looks like a kitchen knife embedded in his skull up to the handle , just to the side of his left eyebrow . according to albuquerque , the knife stopped only when it got stuck in the man 's lower jaw . " he left the hospital yesterday , five days after coming in with a knife in his head , " albuquerque said ."

[Summary-Reference Matching Visualization]

Figure 5: Highlighted matches between generated summary and reference

4.2.4 Example Summary Weighted Approach

For extractive summarization, each sentence in a document is embedded and the document representation was computed as the mean of its sentence vectors. Sentence relevance was then estimated using cosine similarity between each sentence vector and the document vector. A fixed proportion of sentences (40%) with the highest similarity scores is selected, preserving their original order to form the final extractive summary.

Article: *'Comedian Joan Rivers died of "therapeutic complications" during a procedure to evaluate her "voice changes" and stomach reflux, the New York medical examiner said Thursday."The classification of a death as a therapeutic complication means that the death resulted from a predictable complication of medical therapy," the report said.Rivers, 81, died September 4, a week after an appointment at Manhattanś Yorkville Endoscopy clinic.The medical examinerś investigation ruled that the cause of death was "anoxic encephalopathy due to hypoxic arrest," a medical description of brain damage caused by a lack of oxygen.The arrest happened "during laryngoscopy and upper gastrointestinal endoscopy with propofol sedation for evaluation of voice changes and gastroesophageal reflux disease," the medical examinerś officer said.The complication in this case was laryngospasm – a spasm of the vocal cords that makes it difficult to breathe, pathologist and surgeon Bill Lloyd told CNN. The situation was compounded when Rivers went into cardiac arrest."She*

stopped breathing and her heart went into arrhythmia because there wasnt́ sufficient oxygen to maintain the heart muscle," said Lloyd, who was not involved in Riversáutopsy. "The pump, the heart itself, was unable to move fresh oxygenated blood upstream to the brain."At first, Lloyd said, Rivers was unable to bring oxygen into her body due to the vocal cord spasm. After a couple of minutes, her heart started to beat irregularly or completely stopped. Finally, her heart stopped delivering fresh oxygenated blood to her brain, which shut down, according to Lloyd."The unanswered question is why they did not do a combat (tracheotomy)?" he asked. "Youv́e seen in the James Bond movies. You take a ballpoint pen and puncture the thyroid cartilage and create an emergency airway for her. Of course, they would do it with sterile instruments. ... And, after minute two, did anybody say, We need to start doing some CPR for her?"The initial report did not name the doctor or doctors who performed the procedures.Melissa Rivers posted this on Twitter on Friday afternoon: "In response to NYCś Medical examinerś report, we continue to be saddened by our tragic loss. No further comment at this time."Timeline emerges in Joan Riversd́eathWhat happened?Several clinic workers told investigators that the August 28 appointment began with Riversṕersonal throat doctor, Dr. Gwen Korovin, performing a laryngoscopy, which involves using a device to view a patientś vocal folds, a source close to the investigation told CNN last month.Gastroenterologist Dr. Lawrence Cohen, who was the medical director of the clinic until resigning after Riversd́eath, then performed an endoscopy intended to diagnose why she was suffering a sore throat and hoarseness, the source said.Cohen detected something of concern, the source said.Korovin then began a second laryngoscopy to again view Riverś vocal cords, the source said. It was at that time that her vocal cords began to swell, leading to a cutoff of oxygen to her lungs and ultimately to cardiac arrest, according to the source.Korovin was authorized only to observe Cohen, who performed the procedure, since she was not certified by Yorkville Endoscopy clinic, as required by New York health law, the source said.Investigators have found no prior consent form signed by Rivers authorizing a procedure by Korovin, the source said. It was unclear if Rivers had given verbal consent to the biopsy before being sedated.But Korovin denied "performing an unauthorized procedure" before the comedian suffered cardiac arrest, a source close to the doctor told CNN last month.Paramedics rushed Rivers from the clinic to New Yorkś Mount Sinai Hospital a mile away, where she was kept on life support until she died a week later.E!ś F́ashion Policet́o continue without Joan Rivers-Doctor to the starsis well known for helping an impressive list of celebrities with voice trouble. The list of famous patients who have sung her praises include actors Hugh Jackman and Nathan Lane and singers Celine Dion, Lady Gaga and Ariana Grande.walls of Korovinś Manhattan medical office are covered with autographed photos, including from operatic tenor Luciano Pavarotti, Broadway star Barbara Cook and actress-singer Julie Andrews."Iv́e always been fascinated by the human voice and music," Korovin, 55, is quoted telling the New York Daily News in a profile story last year.Korovinś lawyer sent a statement to CNN last month in response to reports that questioned whether her actions played any role in Riversd́eath."Gwen S. Korovin, M.D. is a highly experienced, board certified otolaryngologist. She maintains privileges at one of the cityś most prestigious hospitals. She is respected and admired by her peers in the medical community and she is revered by her patients."As a matter of personal and professional policy, Dr. Korovin does not publicly discuss her patients or their care and treatment. Further, Dr. Korovin is prohibited by state and federal confidentiality laws from discussing her care and treatment of any particular patient."'

Example Summary Weighted Approach:

comedian joan rivers died of " therapeutic complications " during a procedure to evaluate her " voice changes " and stomach reflux , the new york medical examiner said thursday . several clinic workers told investigators that the august 28 appointment began with rivers ' personal throat doctor , dr. gwen korovin , performing a laryngoscopy , which involves using a device to view a patient 's vocal folds , a source close to the investigation told cnn last month . gastroenterologist dr. lawrence cohen , who was the medical director of the clinic until resigning after rivers ' death , then performed an endoscopy intended to diagnose why she was suffering a sore throat and hoarseness , the source said . but korovin denied " performing an unauthorized procedure " before the comedian suffered cardiac arrest , a source close to the doctor told cnn last month .

4.3 Discussion

4.3.1 Topic Modeling Findings

1. **LDA Outperforms LSA:** LDA achieved higher C_V coherence (0.4271 vs 0.3760), indicating more interpretable topics.
2. **Optimal Topic Count:** Lower topic numbers (5 for LDA) produced better coherence, suggesting dominant themes with overlapping content.
3. **BERTopic Granularity:** BERTopic discovered 155 fine-grained topics but with 30.6% outlier documents.

4.3.2 Text Summarization Findings

Based on the Rouge evaluation values obtained, TextRank (Graph Based) is clearly the better method overall.

1. **Much higher Rouge 1:** salient content capturing
2. **Higher Rouge 2:** phrase level coherence.
3. **Higher Rouge L:** structural overlap with references.

4.3.3 Limitations

- Extractive methods cannot paraphrase or generate novel sentences
- Word2Vec trained on limited corpus may miss domain-specific semantics
- BERTopic coherence not directly comparable to LDA/LSA metrics
- Computational cost of hyperparameter search (6+ hours for LDA)

5 Conclusion

This project implemented and evaluated multiple approaches for topic modeling and extractive text summarization on the CNN/Daily Mail dataset (92,579 documents).

Topic Modeling:

- LDA with 5 topics achieved the best coherence (C_V = 0.4271)

- Discovered coherent themes: Sports, Politics, International Affairs, Family/Society, Business
- BERTopic provided fine-grained categorization with 149 topics

Text Summarization:

- Implemented two extractive approaches: TextRank with TF-IDF and Word2Vec with TF-IDF weighting
- TextRank proved to be the better method

References

[1] Hermann, K. M., et al. (2015). Teaching machines to read and comprehend. *NeurIPS*.

[2] Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent Dirichlet Allocation. *JMLR*, 3, 993-1022.

[3] Grootendorst, M. (2022). BERTopic: Neural topic modeling with a class-based TF-IDF procedure. *arXiv:2203.05794*.

[4] Mihalcea, R., & Tarau, P. (2004). TextRank: Bringing order into text. *EMNLP*.

[5] Mikolov, T., et al. (2013). Efficient estimation of word representations in vector space. *arXiv:1301.3781*.